From: **Zoom** no-reply@zoom.us
Subject: Meeting assets for FLEX-DAC team meeting (Thursdays) are ready!
Date: October 29, 2026 at 14:30
To: jflory@asu.edu

zoom

Meeting assets for FLEX-DAC team meeting (Thursdays) are ready!

Meeting summary

Quick recap

This was a project team meeting for a Direct Air Capture (DAC) system integration project with a February deadline. The team members provided introductions and updates on their respective work areas: Jose and Meng are working on systems-level modeling and energy market integration, Avan is handling process modeling and LCA for system comparison, Leonardo is developing CFD simulations for pressure drop analysis, and Marcus is working on CFD simulations for two-phase flows. The discussion focused heavily on the need for mathematical models of the DAC system, particularly energy conversion models that could be integrated into power systems analysis. The team identified that existing models from previous work by Emmy on Aspen Plus simulations would need to be shared with Avan and Jose to help develop the required mathematical expressions. Additional discussions covered the use of potassium carbonate as the preferred carbonate solution, temperature differences for the ladder concept, and the need for experimental calibration data to validate the CFD models. The team agreed to prepare visual presentations with slides for the next meeting to better communicate their progress and milestones.

Next steps

- Justin: Share Emmy's existing process model for the DAC system and Energy Dome analysis with Avan Kumar and Jose.
- Justin: Introduce Avan Kumar to Lindsay Hamblin and ask her to share her models and thesis related to the ladder concept.
- Meng and Jose: Conduct literature review to find mathematical models for the energy conversion processes in the DAC system.
- Avan Kumar: Conduct literature review on process modeling approaches for the DAC system.
- Avan Kumar and Jose: Coordinate and meet (preferably before November 7th) to discuss the DAC model needs for the energy systems analysis.
- Samantha: Measure and provide the geometry and percent openness of the mesh (125 and 25) to the team.
- Samantha: Create custom packets with beads for pressure drop testing and conduct the tests (including a blank test stand measurement).
- Leonardo: Calibrate the CFD model using the upcoming pressure drop data and prepare visual updates for the next meeting.
- All participants: Prepare a few slides for the next meeting, including milestones, context, and next steps.
- Justin: Schedule the next team meeting for Thursday, November 12th, and cancel the meeting on November 13th.
- Avan Kumar: Book a room and meet with Justin in person on Wednesday at 10:30 AM for 1 hour.

Summary

DAC Systems Integration Project Overview

The team held an introductory meeting for a project involving Direct Air Capture (DAC) systems integration into data centers. Justin provided an overview of the project timeline and meeting structure, explaining that team members would give 10-minute updates on their progress every three meetings. Jose presented first, explaining that his team needs mathematical models of the DAC system and components to integrate into their power system analysis, along with electricity price data and information about tariffs, subsidies, and regulations for data centers in selected areas like Arizona.

DAC System Mathematical Modeling

The team discussed the need for a mathematical model of the DAC (Direct Air Capture) system that converts wind energy into stored power. Jose explained they need models for both energy conversion from wind to stored energy and from stored energy back to output power. Justin clarified that while there is some energy storage for maintaining airflow in the front end, the main energy storage involves compressing CO2 into liquid form using renewable energy, which can later be converted back to electrical energy or used for cooling. The team agreed to share existing process models from a previous project with Yvonne, who used Aspen Plus, to help Avan and Jose develop the required mathematical models.

DAC System Development Discussion

Justin discussed the project's advantages over Energy Dome's thermal energy storage approach and recommended pricing out opportunities in Arizona and Texas, particularly considering wind and dry climates. Meng and Jose proposed building a mathematical model by reviewing literature on similar systems and combining components based on the specific configuration, with Meng requesting more system information to locate relevant literature. Avan Kumar presented a flow diagram outlining the DAC system parameters and carbonation processes, asking questions about final CO2 demand, carbonate system selection, and waste heat requirements for liquification.

Carbon Capture Project Progress Update

The team discussed progress on several aspects of their carbon capture project. Justin recommended potassium carbonate as the preferred carbonate solution and mentioned that Lindsay Hamblin's thesis contains detailed information about the ladder concept they're exploring, which requires a 30-35 degree temperature difference for optimal operation. The group reviewed the Energy Dome concept, which uses mechanical compressors to produce liquid CO2 with 75-80% round-trip efficiency, though it has limitations on storage time and requires large domes. Leonardo reported being nearly finished with CFD modeling of the direct air capture system, needing only experimental calibration data including measurements with and without beads in the mesh, and Justin requested that team members prepare visual slides with context and plans for future meetings.

Attendees: Justin Flory (Organizer), Avan kumar (External), Marcus Herrmann, Jose Manuel Dolot, Meng Wu, Samantha Taylor, Leonardo Dayal (External), Avan Kumar (External)

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